

Discovery and Analysis of Complex Patterns

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Introduction

Complex patterns can be observed in many natural phenomena, including super-conductivity, skin pigmentation, swarming, synchronization, etc. We present results from two research directions on large groups of coupled swarming oscillators as well as reaction-diffusion PDEs. In [1], we analyze a swarmalator model on a ring with attractive and repulsive interactions and study its steady state patterns. Moreover, we present a structure preserving method for stabilizing PINNs [2] for solving reaction-diffusion and convection-diffusion PDEs, where the capture of intricate patterns are crucial.

Swarmalator Model

Model Description:

$$\dot{x}_i = \nu + \frac{J_i}{N} \sum_j \sin(x_j - x_i) \cos(\theta_j - \theta_i)$$

$$\dot{\theta}_i = \omega + \frac{K_i}{N} \sum_j \sin(\theta_j - \theta_i) \cos(x_j - x_i)$$

For simplicity, we set $J_i = J = 1$ and draw the phase coupling K from

$$h(K) = p\delta(K - K_p) + q\delta(K - K_n)$$

where $p + q = 1$.

Order Parameters:

$$W_{\pm} = S_{\pm} e^{i\phi_{\pm}} := \frac{1}{N} \sum_{j \leq N} e^{i(x_j \pm \theta_j)} \quad (i = \sqrt{-1}),$$

$$V := \frac{1}{N} \sum_{j \leq N} \langle |\dot{x}_j| \rangle_t.$$

Results:

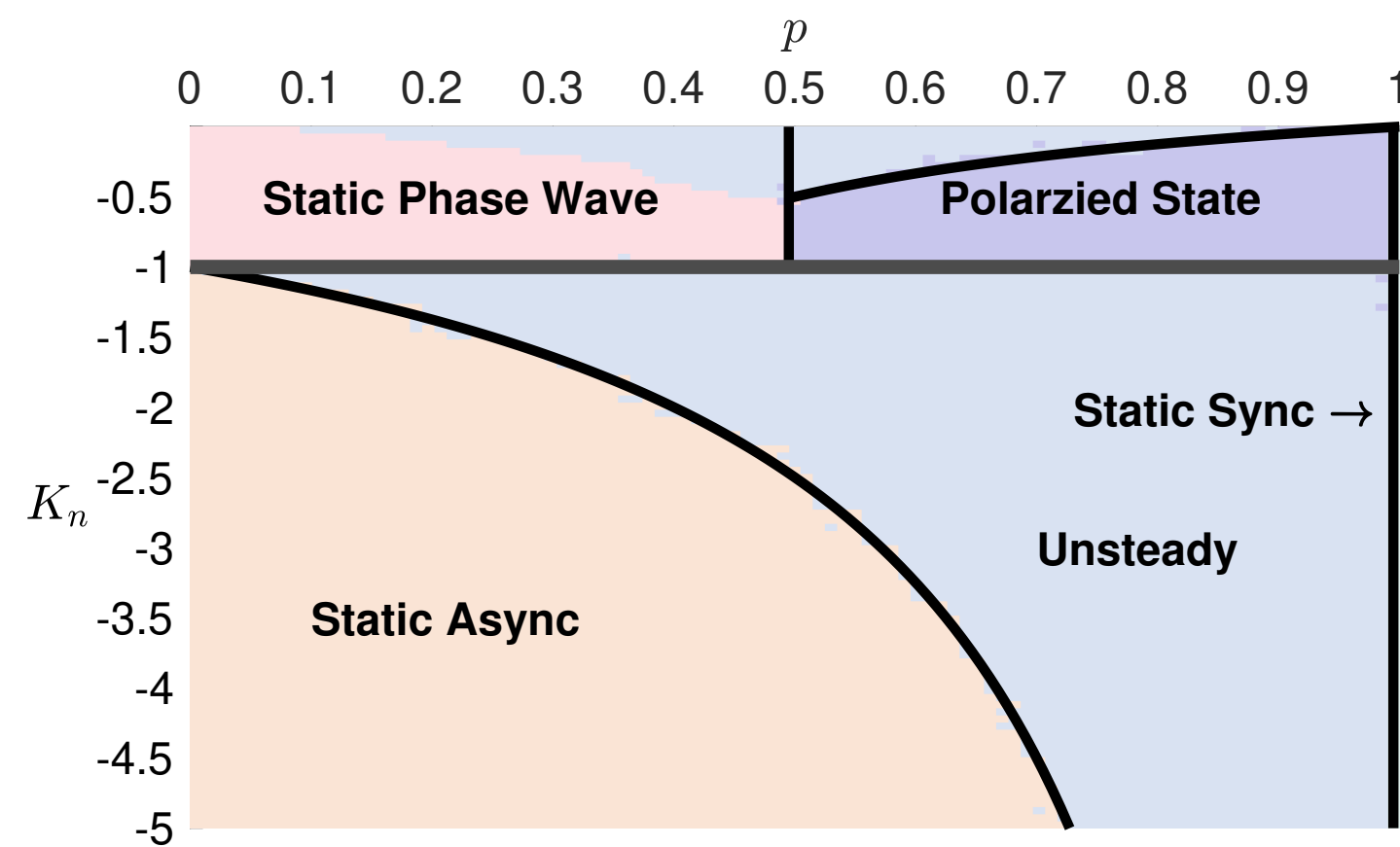


Figure 1. Phase Diagram in (P, K_n) Plane with fixed $K_p=0.5$.

Analysis of Collective States

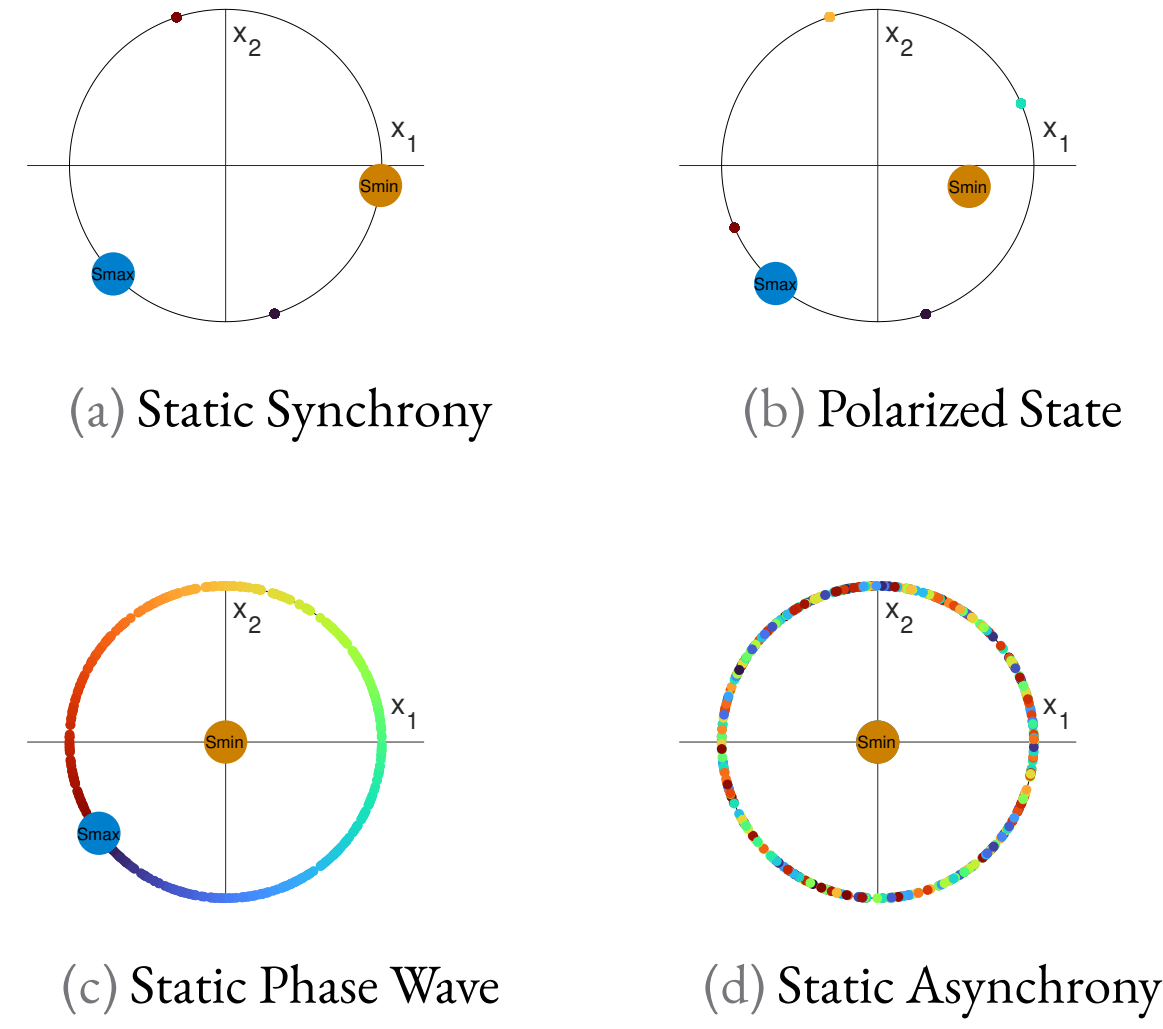


Figure 2. Stationary Collective States.

Other Coupling Distributions:

Single Gaussian Distribution: $h(K) \sim N(\mu, \sigma|K)$.

Mixed Gaussian Distribution: $h(K) \sim pN(K_n, \sigma|K) + (1 - p)N(K_p, \sigma|K)$.

See more results in [1].

Analysis:

- Polarized state :

$$J(qK_p + pK_n) < 0,$$

$$J + K_n > 0,$$

$$p \geq 1/2.$$

- Static state:

$$J > 0,$$

$$K_n, K_p > 0.$$

- Incoherence:

$$p_s = \frac{1 + K_n}{K_n - K_p}.$$

Structure-Preserving PINN

Motivation:

- Possible ill-conditioning of the PDE solution map’s dependence on IC and BC.
- Trying to avoid multi-object optimization problem of baseline PINN [3].

Methodology:

We consider the following transformation

$$\tilde{u}(t, \mathbf{x}) = \psi(t, \mathbf{x}) + \phi(t, \mathbf{x})u_{nn}(t, \mathbf{x}),$$

where ψ and ϕ are smooth functions with the following properties

$$\phi(0, \mathbf{x}) = 0 \quad \text{for } \mathbf{x} \in \bar{\Omega} \quad \text{and} \quad \phi(t, \mathbf{x}) = 0, \quad \text{for } \mathbf{x} \in \partial\Omega;$$

$$\psi(0, \mathbf{x}) = u_0(\mathbf{x}) \quad \text{for } \mathbf{x} \in \bar{\Omega} \quad \text{and} \quad \psi(t, \mathbf{x}) = 0, \quad \text{for } \mathbf{x} \in \partial\Omega.$$

To build in the information of periodicity, we consider the composition

$$u_{nn}(t, \mathbf{x}) = f_{nn}(\mathbf{v}(t, \mathbf{x})), \quad f_{nn} \text{ is a neural network, and } \mathbf{x} \in \mathbb{R},$$

$$\mathbf{v}(t, \mathbf{x}) = \left[t \quad 1 \quad \cos\left(\frac{2\pi}{P}\mathbf{x}\right) \quad \sin\left(\frac{2\pi}{P}\mathbf{x}\right) \quad \cos\left(\frac{2\pi}{P}2\mathbf{x}\right) \quad \sin\left(\frac{2\pi}{P}2\mathbf{x}\right) \cdots \sin\left(\frac{2\pi}{P}m\mathbf{x}\right) \right]^\top.$$

Now the PDE loss is, i.e.

$$\text{Loss}(u_{nn}) = \frac{1}{N_{CL}} \sum_{i=1}^{N_{CL}} |(\tilde{u}_t - \mathcal{P}(\tilde{u}))(\mathbf{t}_i^{CL}, \mathbf{x}_i^{CL})|^2, \quad \text{for } \tilde{u} = \psi + \phi u_{nn}.$$

Results

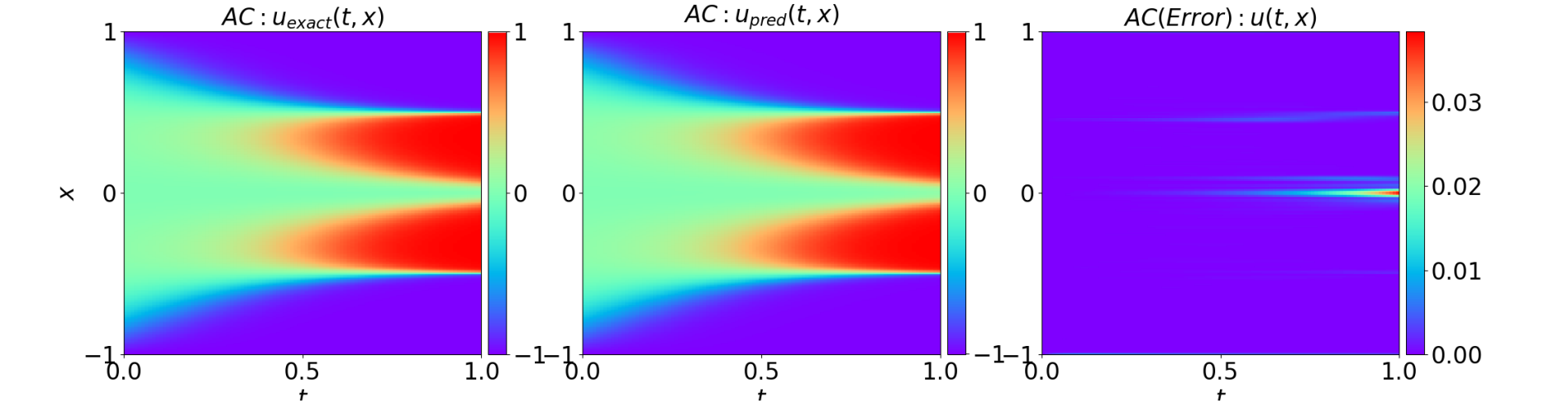


Figure 3. Exact solution of the 1D AC with the prediction and the absolute error difference.

	Allen-Cahn	Baseline PINN	Re-sampling [4]	Our Approach
Relative L_2		9.90e-1	2.33e-2	9.16e-4
Relative L_1		9.90e-1	6.20e-3	4.84e-4
L_∞ norm		9.96e-1	2.64e-1	1.43e-1

Table 1. Comparison of errors in the learned solutions of the AC equation using various PINN.

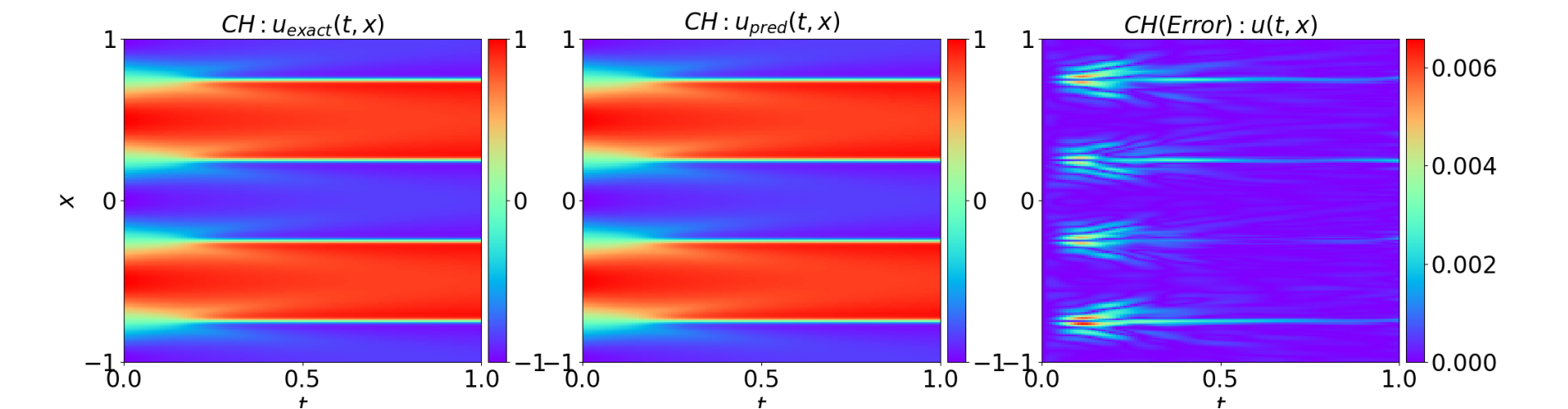


Figure 4. Exact solution of the 1D CH with the prediction and the absolute error difference.

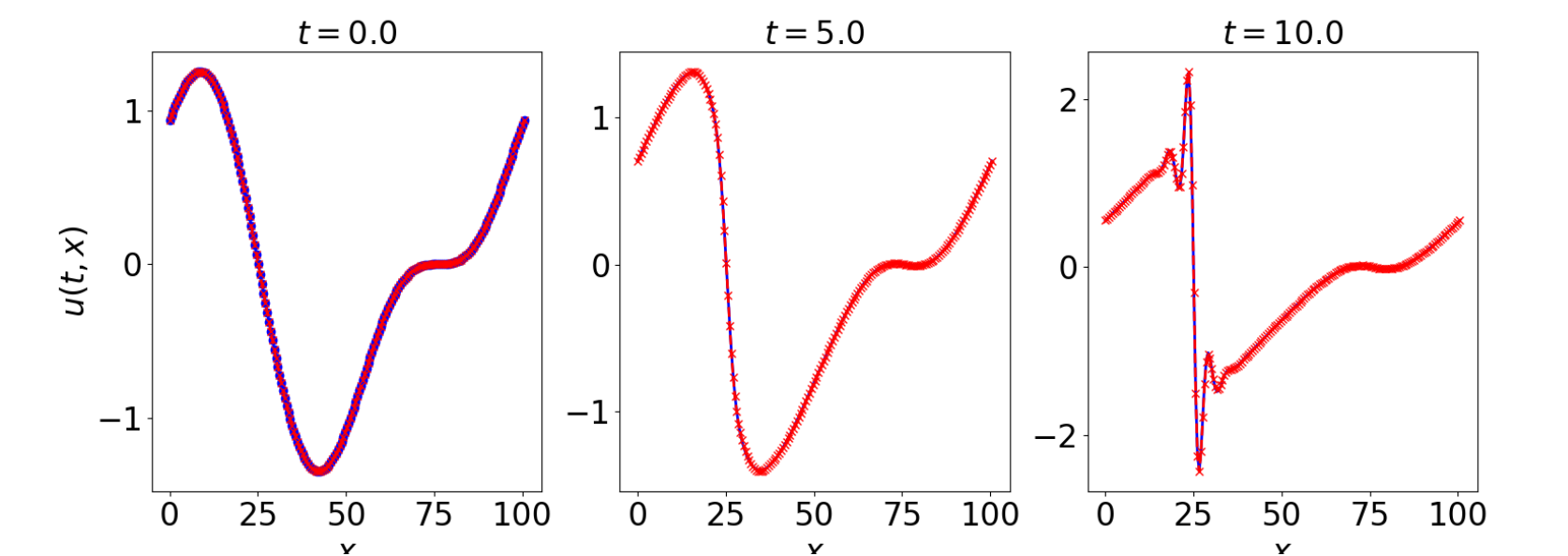


Figure 5. Solutions of the KS equation (4th order PDE).

See more results in [2].

References

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